

COMP460 Machine Learning (Spring 2026)

PROJECT PROPOSAL GUIDELINES

Students are required to prepare a **comprehensive, technically rigorous and research-oriented project proposal**. The proposal must demonstrate **novelty**, clarity of thought and a strong understanding of ML methodologies.

1. Project Topic

Select one topic from the following for your project:

- Project 1: Content Moderation and Toxicity Classification
- Project 2: Text Similarity and Duplicate Detection
- Project 3: Tabular Regression
- Project 4: Time Series
- Project 5: Research Paper Implementation

Details of each project is given in the Project PDF.

2. Team Information

- Maximum **two members** are allowed.
- Provide full names, email addresses and student IDs.
- Clearly define **roles, responsibilities and individual contributions** (e.g., model development, experimentation, documentation, evaluation).

3. Introduction

- Describe the **problem statement** your project aims to solve, including background, context, and existing challenges.
- Explain the **relevance and significance** of the problem within modern ML and real-world applications.
- Highlight the **novelty** of your idea what makes your approach different from conventional methods?

4. Objectives

State **well-defined, measurable and achievable** goals for your project.
Include both:

- **Primary Objectives** (core outcomes you aim to achieve)
- **Secondary Objectives** (extensions, enhancements, or exploratory components)

5. Methodology

- a. Describe in detail the **ML techniques, algorithms, architectures and models** you plan to utilize.
- b. Explain **how each method contributes** to solving the problem.
- c. Present a **technical workflow** such as:
 - data ingestion & preprocessing

- feature engineering
- model selection
- training procedures
- hyperparameter tuning

d. Highlight any **novel techniques or custom algorithms** you plan to implement.

6. Dataset

a. Identify all **datasets** you will use for training, validation and testing.

b. Include details regarding:

- Source
- Size and number of samples
- Data format (images, text, audio, tabular, etc.)

c. Discuss required **preprocessing**, such as normalization, augmentation, cleaning or balancing.

d. If creating your own dataset, include the **data acquisition plan** and feasibility.

7. Tools and Technologies

1. List all **Python libraries, frameworks** (e.g., PyTorch, scikit-learn, TensorFlow, Kaggle, Colab).
2. Justify each tool choice by connecting it to your **methodology, model type, experiment needs and resource constraints**.

8. Milestones and Timeline

Provide a **well-structured Gantt-style timeline** covering the full project cycle, with expected deadlines for:

- Literature review
- Problem understanding
- Data collection/preprocessing
- Model architecture design
- Model training & experimentation
- Evaluation & comparison
- Results analysis
- Documentation & final submission

Ensure the timeline is **realistic and logically sequenced**.

9. Evaluation Metrics

- a. Define the **quantitative metrics** you will use (e.g., Accuracy, F1-Score etc).
- b. Explain why these metrics are appropriate based on your **problem type** (classification, regression, etc).
- c. Discuss any **baseline models** you will use for comparison to demonstrate improvement.

10. Novelty & Innovation

Explicitly describe:

- What makes your project **unique**
- Which part involves **innovation or improvement** over existing techniques
- Any **creative components, optimization strategies or hybrid models** you plan to explore

11. Resources

List any additional sources of inspiration, including:

- Research papers
- Books
- Online datasets
- Technical blogs or documentation

12. References

Follow a consistent citation style (APA/IEEE). Cite all:

- Papers
- Articles
- Datasets
- Tools/libraries documentation

13. Submission Requirements

- Submit a **properly formatted PDF or DOCX** containing all the above sections.
- Submission Platform: **LMS**
- Late submissions may be penalized according to course policy.

14. Deadlines:

Group Creation: **5th March**

Project Selection: **5th march (After that you'll not be able to edit the sheet)**

Deliverable 1 (30% project completion): **31st march**

Deliverable 2 (70% project completion): **21st march**

Final Deliverable (100% project completion): **12 may**

Feedback:

For any queries regarding the project. Reach out to

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